

30. PROPERTIES OF THE PERITONEUM

DURATION : 04:49

STUDY SHEET

COURSE SUMMARY

In this video, explore the often underestimated peritoneum as an active and dynamic organ. The teaching highlights its essential function in the mobility of intraperitoneal fluid, its role in embryological integration, and its involvement in molecular exchanges within the body. The concepts of cellular continuity and homeostasis are explored in depth, illustrating how interactions at the peritoneal level can trigger distant immune responses, such as allergic reactions. Valuable insights into the links between the peritoneum and seemingly unrelated symptoms, such as asthma, make this video crucial for any somatic therapy professional.

PROPERTIES OF THE PERITONEUM

The peritoneum plays an essential role in the **passive** and constant **mobility** of the **intraperitoneal fluid**. It also participates in walking, especially in cases of peritoneal fixation. The upper and lower limbs can be considered an **expansion** of this peritoneal sac. What happens in this region can reflect development in other areas, illustrating **cellular, tissue, molecular, and even energetic continuity**.

It is crucial to consider the peritoneum as an **active organ**. It possesses **immune protection** capabilities and releases **tissue hormones**, such as **prostaglandins**, depending on secretion and absorption. This means it attempts to maintain a constant fluid volume, close to that of the **cerebrospinal fluid**.

The integration of the **external cellulome** into an **internal cellulome** unfolds in a dynamic similar to that of the integration of the **neural tube**. This can be compared to the integration of **amniotic fluid** into the neural tube and **extra-psolomic fluid** into the peritoneal system. This process concludes around the **28th day**, with a peritoneal volume that seeks to maintain **constancy** to ensure a homeostatic hydrostatic balance, both cerebrally and peritoneally.

The peritoneum is also involved in molecular exchanges. It contains **mesenchymal cells** which, beyond the precordial plate, become the **pericardium** and the **septum transversum**. These cells are integrated by the global movement of the embryo, particularly during heart development, which interacts with the **vitelline vesicle**. This integration process occurs between the **22nd** and **28th day**.

The peritoneum is an organ rich in **arterioles** that exchange **osmotically** with blood plasma, providing exudates that inform about the **intracellular vegetative state**. It is constantly informed of the digestive and metabolic state of the surrounding organs through its interstitial fluid.

Mesothelial-type cells, such as **macrophages**, **polymorphonuclear cells**, and **mast cells**, play a crucial role in defence reactions at the peritoneal level. Mast cells, sensitive to **immunoglobulin E (IgE)**, are involved in allergic and asthmatic reactions. For example, an infectious aggression at the peritoneal level can lead to **mast cell degranulation** which manifests remotely, such as an asthmatic reaction.

A poor immune barrier at the peritoneal membrane can allow undesirable macromolecules to pass, leading to allergic reactions. For example, a milk allergy can cause an asthmatic reaction, even if the infectious agent is initially peritoneal.

The digestive tract, considered an external tissue, can also express problems through symptoms such as asthma or bronchitis, in response to allergenic agents. Treatments, such as **bronchodilators** or **antibiotics**, can relieve symptoms but do not always resolve the underlying cause, which may simply be an allergy to a food, such as a strawberry.